Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/28/2008
PI Name:	Frenklach, Michael Y. Ph.D.		
Project Title:	Microgravity Production of Nanoparticles of Novel Materials using Plasma Synthesis		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	12/13/2000	**End Date:**	10/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Frenklach, Michael		
COI Name (Institution):	Fernandez-Pello, Carlos (University of California, Berkeley)		
Grant/Contract No.:	NCC3-833		
Performance Goal No.:			
Performance Goal Text:			

The proposed research program is to study the formation in reduced gravity of high quality nanoparticulate of novel materials using plasma synthesis. Particular emphasis will be placed on the production of powders of non-oxide materials like diamond, SiC, SiN, c-BN, etc. The goal of the proposed study is to investigate the effect of gravity on plasma synthesis of these materials, and to determine how the microgravity synthesis can improve the quality and yield of the nanoparticles.

A plasma reactor, operating in the diffusion flame mode, will be designed and built. The reactor will be designed to operate from low to near atmospheric pressure, keeping in mind that it will be used in ground based microgravity facilities. The expectation is that the larger residence times and better nucleate uniformity obtained in microgravity, along with higher pressures, will result in powder of high yield and quality. Normal gravity experiments will be conducted to determine the performance characteristics of the reactor, i.e., powder yield and quality, and to obtain data

Task Description:	for comparison with the microgravity experiments. The initial experiments will be conducted with silicon carbide powder, which has similar synthesis to diamond but requires a somewhat simpler system. Upon successful accomplishment of this task, the experiments on diamond powder and other materials of interest will be performed. It is expected that the reduced gravity will aid in the understanding of the controlling mechanisms of plasma synthesis, and will increase the yield and quality of the synthesized powder. These materials have properties of interest in several industrial applications, such as high temperature load bearings or high speed metal machining. Furthermore, because of the nanometer size of the particulate produced in this process, they have specific application in the fabrication of MEMS-based combustion systems, and in the development and growth of nanosystems and nanostructures of these materials. These are rapidly advancing research areas, and there is a great need for high quality nanoparticles of different materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is expected that the reduced gravity will aid in the understanding of the controlling mechanisms of plasma synthesis, and will increase the yield, and quality of the synthesized powder. These materials have properties of interest in several industrial applications, such as high temperature load bearings or high speed metal machining. Furthermore, because of the nano-meter size of the particulate produced in this process, they have specific application in the fabrication of MEMS based combustion systems, and in the development and growth of nano-systems and nano-structures of these materials. These are rapidly advancing research areas, and there is a great need for high quality nanoparticles of different materials.
Task Progress:	To accomplish the stated objectives, a plasma reactor, operating in the diffusion flame mode, was designed and built at the University of California at Berkeley. The Berkeley team combined the experience in plasma synthesis of diamond and other non-oxide materials and in premixed, diffusion and condensed phase combustion in microgravity. The reactor was designed keeping in mind that it will be used in ground based microgravity facilities, to operate from low to near atmospheric pressure. The expectation was that the larger residence times and better nucleate uniformity obtained in microgravity, along with higher pressures, will result in powder of high yield and quality. Normal gravity experiments were conducted to determine the performance characteristics of the reactor, i.e., powder yield and quality, and to obtain data for comparison with the microgravity experiments.
Bibliography Type:	Description: (Last Updated: 02/27/2004)